SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Vulnerability Detection

Course Code:	CSA5803	**Course Title:**	DEVSECOPS ENGINEERING AND APPLICATION SECURITY
CO Assessed:	CO3 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 3
Weightage:	40% of CIA	**Total Marks:**	100
CO 3 Statement:	Vulnerability Detection, Severity Analysis, Security Verification Workflow Simulation and Performance Evaluation		
Student Name:		**Reg. No.:**	
Section / Batch:		**Date:**	

Instructions to Students

- Perform vulnerability detection and classify the identified vulnerabilities based on their severity levels.
- Conduct severity analysis using appropriate metrics such as CVSS and document the impact of each vulnerability.
- Design and simulate a Security Verification Workflow covering detection, remediation, re-testing, and verification.
- Evaluate security and system performance before and after mitigation, and include relevant results, screenshots, and observations.

Question Type	Focus Area	Questions	Marks
Vulnerability Detection	Conceptual Understanding	Q1	Q1 –100
	GRAND TOTAL:	**100 Marks**	

Code of Conduct

I _____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand